Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Getting Started

- Course Introduction
  Video • 1 min
- Course Overview
  Reading • 2 min
- Syllabus
  Reading • 10 min
- Meet Your Instructor
  Video • 1 min
- Meet Your Fellow Learners
  Discussion Prompt • 10 min
- Academic Integrity
  Reading • 1 min

Lesson 1: Introduction to Linear Algebra

- Introduction to Machine Learning
  Reading • 25 min
- Introduction to Linear Algebra
  Reading • 4 min
- Why Linear Algebra and Mathematics?
  Reading • 2 min
- Notation
  Reading • 5 min

Lesson 2: Systems of Linear Equations

- Foundational Concepts of Systems of Linear Equations
  Reading • 20 min
- [H5P] System of Linear Equations
  Ungraded App Item • 20 min
- Solved Example and Recommended Resources
  Reading • 20 min
- Check Your Knowledge: System of Linear Equations
  Graded Assignment • Grade: 100%

Lesson 3: Matrices

- **Matrices and Matrix Operations**
  Reading • 20 min
- Matrices
  Video • 8 min
- Helpful Resources and Solved Examples
  Reading • 10 min
- Check Your Knowledge: Matrices
  Graded Assignment • 16 min

Lesson 4: Vector Spaces

- Foundations of Vector Spaces: Operations and Subspaces
  Reading • 20 min
- Vector Space
  Video • 7 min
- Vector Space Properties
  Reading • 10 min
- Subspace
  Reading • 1 min
- Check Your Knowledge: Vector Spaces
  Graded Assignment • 14 min

Lesson 5: Orthogonal Complement and Projection

- Introduction to Orthogonal Complement
  Reading • 20 min
- Introduction to Orthogonal Projections
  Reading • 40 min

Matrices and Matrix Operations

0:16 / 1:10 2x

Read Section 2.2: “Matrices” from Mathematics for Machine Learning by Marc Peter Deisenroth, A. Aldo Faisal, and Cheng Soon Ong.

In this lesson, you will be introduced to matrices and various matrix operations. You will learn about matrix addition and multiplication, the identity matrix, inverse and transpose operations, symmetric matrices, and multiplication by a scalar. Understanding these concepts is essential for effectively manipulating and transforming data in machine learning algorithms.

As you read, consider the following questions:

- How do matrices provide a compact and efficient way to represent systems of linear equations?
- How do matrix operations facilitate data transformation and manipulation in machine learning?
- Why is understanding matrix inversion and transposition important for solving linear equations in machine learning models?
- How can symmetric matrices and the identity matrix be utilized in simplifying complex calculations and improving computational efficiency?

By the end of this lesson, you should have a solid grasp of basic matrix operations and how they apply to the field of machine learning.

Go to next item ✓ Completed

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